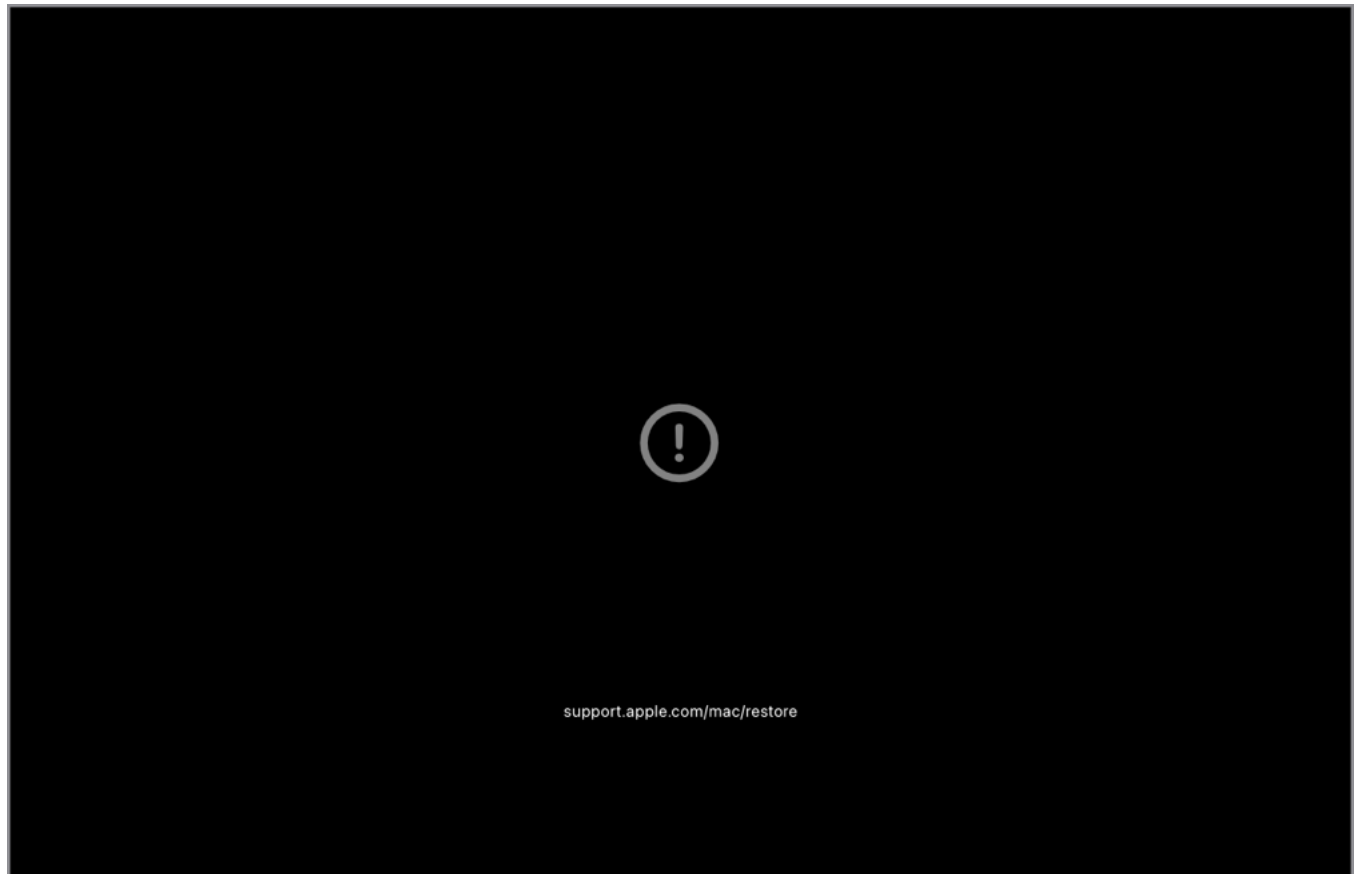


If your Mac starts up with an exclamation mark in a circle

The firmware of your Mac needs to be revived.

March 27, 2025



In very rare circumstances, such as when a power failure interrupts macOS installation, your Mac might start up to a circled exclamation mark. This means that the firmware stored in your computer's memory needs to be revived or restored.

[How to revive or restore Mac firmware](#)

Need more help?

Tell us more about what's happening and we'll suggest what you can do

How to identify the DFU port on Mac

Find out how to locate the device firmware update port on your Mac.

March 13, 2026

For certain tasks, such as [reviving or restoring firmware](#) or [installing macOS on an external storage device](#), you need to know the location of your computer’s DFU (device firmware update) port. Only a [Mac with Apple silicon](#) or [Mac with the Apple T2 Security Chip](#) has this DFU port, which looks identical to other ports that accept a type USB-C connector.

DFU port locations

With some exceptions, the DFU port is the type USB-C port furthest to the left (leftmost) or furthest to the right (rightmost) when you’re facing the ports. Some examples are pictured below the table.

Mac		DFU port location
MacBook Neo	All models	• The leftmost USB-C port when you’re facing the left-hand side of the Mac
MacBook Pro	Models with Apple silicon	• 14-inch MacBook Pro with M4 or M5 chip (excluding Pro or Max versions): The rightmost USB-C port when you’re facing the left-hand side of the Mac • All other models: The leftmost USB-C port when you’re facing the left-hand side of the Mac
	Models with T2 chip	• The rightmost USB-C port when you’re facing the left-hand side of the Mac
	Models	• MacBook Air introduced in 2025 or later: The rightmost USB-C port when you’re facing the left-

MacBook Air	with Apple silicon	hand side of the Mac • All other models: The leftmost USB-C port when you're facing the left-hand side of the Mac
	Models with T2 chip	• The rightmost USB-C port when you're facing the left-hand side of the Mac
iMac	All models	• The rightmost USB-C port when you're facing the back of the Mac
Mac mini	Models with Apple silicon	• Mac mini (2024): The middle USB-C port when you're facing the back of the Mac • All other models: The leftmost USB-C port when you're facing the back of the Mac
	Models with T2 chip	• The rightmost USB-C port when you're facing the back of the Mac
Mac Studio	All models	• The rightmost USB-C port when you're facing the back of the Mac
Mac Pro	Models with tower enclosure	• On the top of the Mac, the USB-C port furthest from the power button
	Models with rack enclosure	• On the front of the Mac, the USB-C port closest to the power button

DFU port examples

These examples show the DFU port on several Mac models. The location on your Mac model may differ. Consult the table above for details.

The leftmost USB-C port on a model of MacBook Pro:



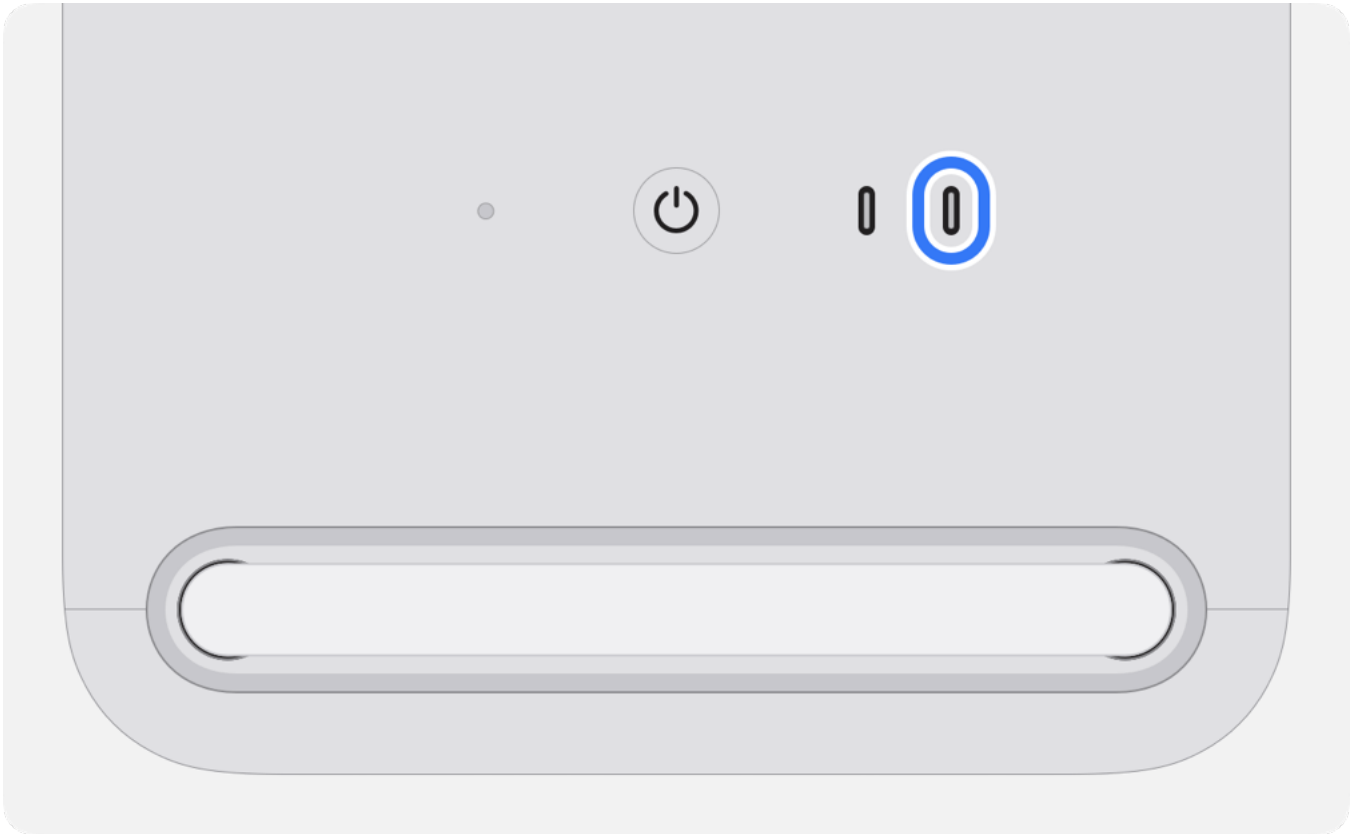
The rightmost USB-C port on a model of MacBook Air with T2 chip:



The rightmost USB-C port on a model of iMac:



The USB-C port furthest from the power button on a Mac Pro with tower enclosure:



How to revive or restore Mac firmware

Rarely, a Mac with Apple silicon or the Apple T2 Security Chip may experience an issue that requires its firmware to be revived or restored by another Mac.

May 05, 2026

When to revive or restore

If the firmware stored in its memory needs to be revived or restored, a Mac with Apple silicon or the Apple T2 Security Chip may experience issues such as the following. These can occur in certain rare circumstances, such as after a power failure that interrupts macOS installation.

- Starts up to an [exclamation point in a circle](#)
- Turns on but starts up to a blank screen that is not resolved by other [solutions for a blank screen](#)
- Shows the [status indicator light pattern](#) for firmware recovery mode
- Experiences a macOS installation issue not resolved by other [solutions for macOS installation errors](#)

What you'll need to revive or restore

1. The affected Mac, which is the [Mac with Apple silicon](#) or [Mac with the Apple T2 Security Chip](#) that you're reviving or restoring. Other Mac models don't apply.

2. Another Mac of any model using macOS Sonoma 14 or later. You will use this Mac to revive or restore the affected Mac. [Find out which macOS your Mac is using](#).



3. A USB-C to USB-C cable that supports data and charging, such as the Apple USB-C Charge Cable included with some Apple products. It works with [ports on Mac](#) that accept a type USB-C

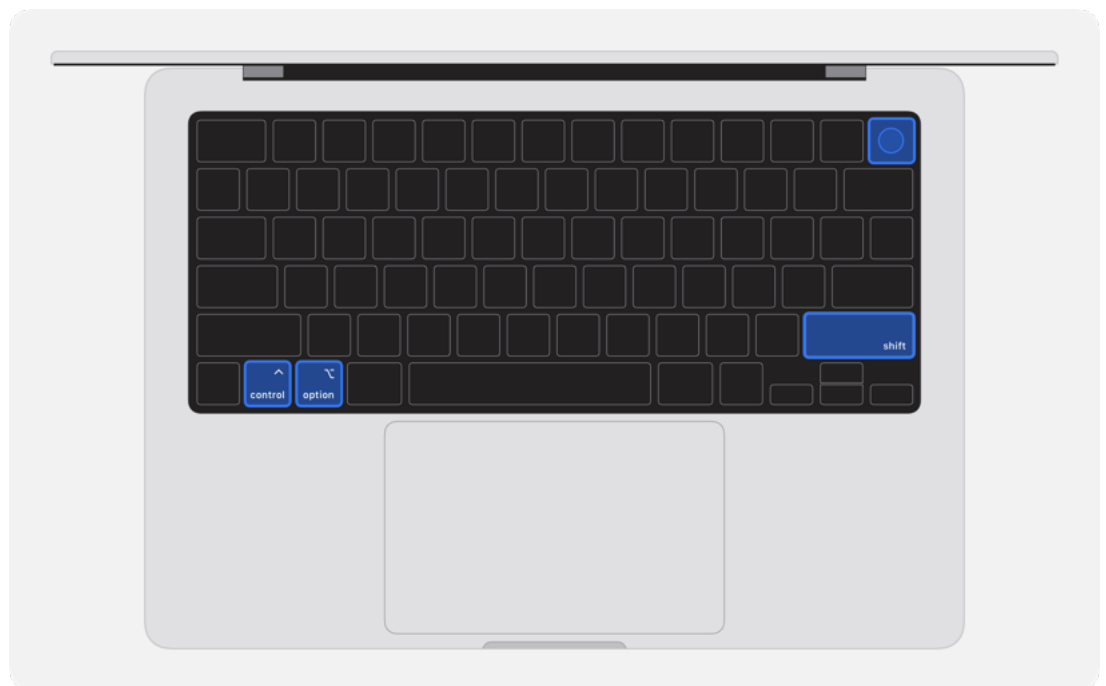
connector. *Don't use a Thunderbolt 3*  *cable.*

How to prepare a laptop to revive or restore

You will use the USB-C cable to connect the two Mac computers, then enter DFU (device firmware update) mode on the affected Mac.

1. Make sure that both Mac computers are connected to power.
2. On the affected Mac laptop:
 - Plug the USB-C cable into the DFU port. [Find out how to identify the DFU port](#).
 - Make sure nothing is plugged into any other USB port, except as needed to connect your Mac to power. If your Mac has a MagSafe port, use that port for power instead of a USB port.
3. On the other Mac:
 - Plug the other end of the USB-C cable into any USB-C port.
 - Make sure this Mac is turned on and connected to the internet.*
4. On the affected Mac laptop, enter DFU mode:

1. Press and hold the power (Touch ID) button for up to 10 seconds until the Mac turns off. If the Mac turns on instead, repeat this step.
2. Press and release the power button, then immediately press and hold all four of these together on the built-in keyboard:
 - Control $\wedge$ on the left-hand side of the keyboard
 - Option $_$ on the left-hand side of the keyboard
 - Shift $_$ on the right-hand side of the keyboard
 - Power button



3. Mac laptop with Apple silicon:

1. Keep holding all four keys for about 10 seconds, then release all keys — except the power button.
2. Keep holding the power button for up to 10 more seconds, until the other Mac shows a DFU window in the Finder. If it first shows an alert asking you to allow the accessory to

connect, release the power button and click Allow.

4. Mac laptop with T2 chip: Keep holding all four keys for about three seconds, until the other Mac shows a DFU window in the Finder. If it first shows an alert asking you to allow the accessory to connect, release all of the keys and click Allow.
5. The affected Mac will now be in DFU mode and should be showing a blank screen. Follow the steps to revive or restore.

How to prepare a desktop computer to revive or restore

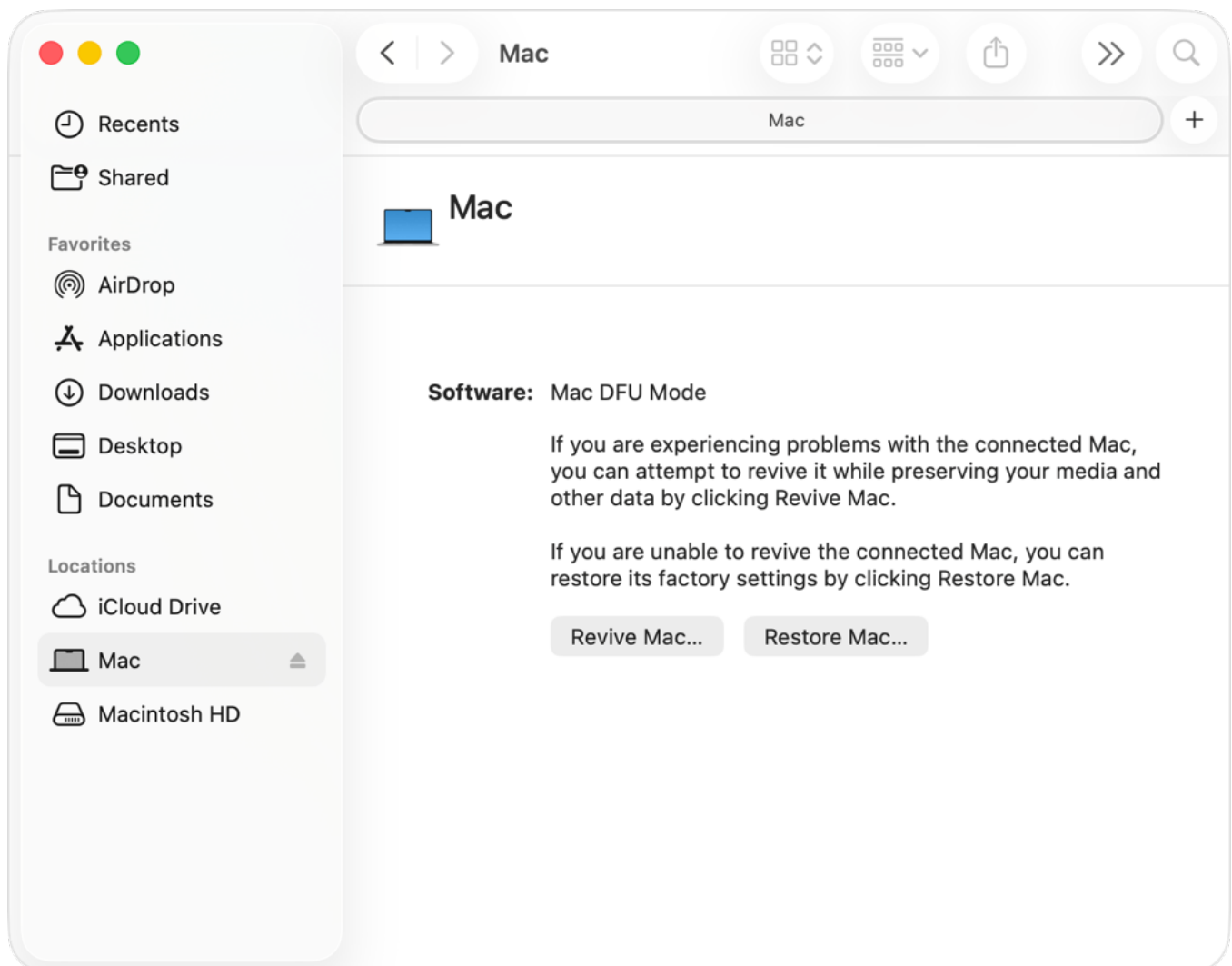
You will use the USB-C cable to connect the two Mac computers, then enter DFU (device firmware update) mode on the affected Mac.

1. On the affected Mac:
 - Plug the USB-C cable into the DFU port. [Find out how to identify the DFU port.](#)
 - Make sure nothing is plugged into any other USB port.
2. On the other Mac:
 - Plug the other end of the USB-C cable into any USB-C port.
 - Make sure this Mac is turned on and connected to a power source and the internet.*
3. On the affected Mac, enter DFU mode:
 1. Unplug the Mac from its power source.
 2. Press and hold the power button.

3. Keep holding the power button while plugging the Mac into power and continue holding it for up to 10 seconds until the other Mac shows a DFU window in the Finder. If it first shows an alert asking you to allow the accessory to connect, release the power button and click Allow.
4. The affected Mac will now be in DFU mode and should be showing a blank screen. Follow the steps to revive or restore.

How to revive or restore

After preparing your computers to revive or restore, the Finder on the other Mac should show a DFU window. The DFU window shows "Mac" selected in the sidebar and "Mac DFU Mode" on the right. Don't see the DFU window?



Revive

Try reviving first. It can be faster than restoring, and it doesn't erase your Mac.

1. From the DFU window in the Finder, click Revive Mac. Then click Continue to confirm.
2. If this Mac is using an earlier version of macOS than the Mac being revived, you might be asked to update the software on this Mac before you can continue.
3. A progress bar in this window will show that the revive is underway. When the revive is complete, the affected Mac restarts automatically. If it shuts down instead, press the power button to turn it on.
4. If asked, select the volume to recover (such as Macintosh HD), then click Next.
5. If asked, select a user you know the password for, then enter that user's login password. Click Next, then click Restart.
6. If reviving a Mac with Apple silicon, the revived Mac might [load startup options](#), which shows your startup disk (such as Macintosh HD) and Options with a gear icon. Select your startup disk, then click the Continue button that appears below it.
7. The revived Mac will finish starting up, then the process will be complete.

Restore

Try reviving your Mac first. If your Mac can't be revived, follow these steps to erase it and restore it to factory settings.

1. From the DFU window in the Finder, click Restore Mac. Then click Restore and Update to confirm.
2. If this Mac is using an earlier version of macOS than the Mac being restored, you might be asked to update the software on this Mac before you can continue.
3. A progress bar in this window will show that the restore is underway. When the restore is complete, the affected Mac restarts automatically. If it shuts down instead, press the power button to turn it on.
4. If asked, select a Wi-Fi network or attach a network cable.
5. Mac with Apple silicon:
 1. If asked, sign in to the Apple Account previously used with this restored Mac.
 2. When the setup assistant opens, use it to finish setting up your Mac.
6. Mac with T2 chip:
 1. The restored Mac will show a spinning globe as it starts up from macOS Recovery over the internet. Choose your language when asked.
 2. If asked, sign in to the Apple Account previously used with this restored Mac. Then click Exit to Recovery.
 3. When you see the list of utilities in Recovery, choose the option to [install macOS](#).
 4. After macOS installation, the Mac restarts to a setup assistant. Use it to finish setting up your Mac.

If you can't see a DFU window in the Finder

After preparing your computers to revive or restore, if you don't see a DFU window in the Finder, try these solutions:

- From the Finder menu in the menu bar, choose Settings. Click Sidebar at the top of the settings window, then make sure that "CDs, DVDs, and iOS Devices" is selected.
- Look for "Mac" in the Locations section of the sidebar of any Finder window. Select it, if present.
- Disconnect the USB-C cable from the affected Mac, then press and hold the power button for up to 10 seconds to turn off the affected Mac. Reconnect the USB-C cable to the [DFU port](#), make sure nothing is plugged into any other USB port, then try again to enter DFU mode. Proper timing of key presses is important.
- Try a different USB-C cable. It must support both data and charging.

Need more help?

Tell us more about what's happening and we'll suggest what you can do next.

[Get suggestions](#)

* If using a web proxy or firewall, it must allow network traffic from your Mac to Apple's network, 17.0.0.0/8. If you're not sure, consult your router manual or internet provider. Find out more about [using Apple products on enterprise networks](#).